Test log

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A. Automated tests: run in the development environment

Command	`pytest -v`
Environment	Linux, Python 3.13.16, pytest 9.1.1
Date	17 October 2026
Result	**17 passed in 16.52s**

#	Test	Result
1	`test_valid_inference_returns_all_fields`	PASSED
2	`test_missing_field_is_rejected`	PASSED
3	`test_wrong_type_is_rejected`	PASSED
4	`test_range_error_is_rejected`	PASSED
5	`test_unknown_season_and_extra_field_are_rejected`	PASSED
6	`test_dataset_recreates_same_fingerprint`	PASSED
7	`test_full_pipeline_reproduces_metrics`	PASSED
8	`test_dataset_has_the_deliberate_problems`	PASSED
9	`test_targets_are_never_inputs`	PASSED
10	`test_schema_check_rejects_missing_column`	PASSED
11	`test_wrong_seed_is_rejected`	PASSED
12	`test_cli_invalid_record_exits_with_error`	PASSED
13	`test_cli_valid_record_prints_json`	PASSED
14	`test_api_predict_returns_expected_response`	PASSED
15	`test_api_invalid_request_returns_422_with_field_names`	PASSED
16	`test_api_rejects_non_json_body`	PASSED
17	`test_api_health_and_model_info_show_code`	PASSED

The tests cover valid inference, missing field, wrong type, range error, unknown season and extra field, reproducibility (dataset fingerprint and full pipeline), the deliberate data problems, leakage, wrong seed, command-line success and failure, and four API checks including a 422 response.

B. Automated tests: run on my own computer

Operating system and version	Windows 11 Pro, version 21H2 (OS build 22000.2538)
Python version	Python 3.14.7
Date and time	18 October 2026, about 16:14 (Africa/Juba)
Command	`pytest -q`
Result line (for example “17 passed in 20s”)	**17 passed, 1 warning in 44.72s** (the warning is a Starlette/httpx deprecation notice)
Code commit hash	`45907a7903dcc070ed1fc265db4fe439047e23cd`

C. Manual checks (fill in after doing them)

Check	Done	Notes
python run_all.py twice gives the same SHA-256 fingerprint		
python predict.py with samples/valid.json prints a prediction and AI-A2-8259	Yes	Yield 12.3615 t, not_high, cluster 5

python predict.py with samples/invalid.json prints an error	Yes	invalid_input, 4 fields named
uvicorn: /health and /model-info respond; /docs opens		
POST /predict: valid gives 200, invalid gives 422		

Section A was produced by an AI assistant in its own environment and is labelled as such. Section B and the manual checks marked Yes in section C were done on my own computer; the other checks in C are still to be completed.